Evaluating Financial AI

A Framework for Benchmarking and Governing LLMs in Finance

Based on: Evaluation and Benchmarking Suite for Financial LLMs and Agents

SecureFinAI Lab, Columbia University (2026)

01

Local Context

The local context of AI in Australian regulated financial environments

TBC

02

Why Financial AI Requires a Different Operating Model

Understanding the unique risks of deploying AI in regulated financial environments

THE PROBLEM

Financial AI introduces new risk categories

Financial AI must be evaluated differently from consumer AI.

Model Risk (Financial Reasoning)

Interpreting earnings reports and SEC filings

Analysing financial statements

Numerical reasoning in investment analysis

Errors affect risk models and trading decisions

Operational Risk (Autonomous Agents)

Retrieving market data autonomously

Analysing financial news

Generating investment insights

Non-deterministic behaviour requires monitoring

Regulatory & Compliance Risk

Training data provenance is unclear

Licensing restrictions are unknown

Outputs cannot be audited by regulators

Governance Gap

No standard for benchmarking financial AI

No framework for evaluating agents in production

No widely adopted transparency standard

THE PROBLEM

The core challenge

Financial institutions deploying GenAI face three critical questions:

Which models can we trust?

How do we evaluate financial reasoning capability and accuracy?

How do we safely operate AI agents?

How do we monitor autonomous systems in production?

How do we meet audit requirements?

How do we ensure regulatory compliance and model transparency?

This presentation outlines a framework addressing all three.

03

The Financial AI Lifecycle

From experimentation to governed deployment

LIFECYCLE

Three stages of financial LLM maturity

Financial AI is evolving through three maturity stages.

2023

Exploration

Early financial LLM development.

BloombergGPT, FinGPT

Can LLMs understand financial language?

Open FinLLM Leaderboard

2024

Readiness

Shift to deployment readiness.

Systematic benchmarking

Financial datasets & evaluation

Emergence of financial agents

2025

Governance

Safe and compliant deployment.

AI governance frameworks

Operational evaluation of agents

Model transparency standards

04

Benchmarking Financial AI

Standardised evaluation across financial tasks

BENCHMARKING

The Open FinLLM Leaderboard

Standardised evaluation across 42 financial datasets, organised into three capability areas.

Understanding Financial Language

Information extraction

Sentiment analysis

Classification

Financial Reasoning

Answering questions about financial data

Summarising earnings reports

Analysing regulatory filings

Financial Decision Tasks

Credit risk scoring

Market forecasting

Trading and investment simulations

Hosted on HuggingFace in collaboration with Linux Foundation, PyTorch Foundation, and SecureFinAI Lab. Models ranked by both performance and openness classification.

05

From Models to Agents

Financial AI agents introduce new capabilities and new risks

FINANCIAL AGENTS

Agents change the risk profile

Financial AI is moving from static models to autonomous agents.

New agent capabilities

Multi-step reasoning across financial data
Tool usage for data retrieval and analysis
Autonomous decision-making
Real-time financial data retrieval

Traditional LLM evaluation measures answers.
Agent evaluation must measure behaviour.

Agentic FinSearch

Autonomous research agent retrieving financial data and analysing news sentiment

Financial Tutor

AI-powered financial education: CFA prep and financial training at scale

FinSight Agent

Multi-agent earnings call analysis with self-evaluation and guardrails

06

Governing Financial AI Systems

Operational evaluation and model transparency

GOVERNANCE

AgentOps: Evaluating AI agents in practice

Two layers of governance for the AI agent lifecycle.

1 Development Governance (Inner Loop)

"Glass box" evaluation of agent behaviour during development.

- Reasoning validation
- Tool usage testing
- Context adherence
- Safety guardrails

2 Operational Governance (Outer Loop)

Continuous evaluation of AI systems in production.

- Decision tracing
- Audit logs
- LLM-as-a-Judge safety scoring
- Human-in-the-loop escalation

> Production failures are fed back into training datasets to continuously improve reliability.

GOVERNANCE

Model Openness Framework

Avoiding "open-washing" in AI model procurement.

Many models claim to be open but hide training datasets, licensing terms, and development processes.

CLASS	LEVEL OF OPENNESS	WHAT'S SHARED
Class III	Open Model	Partially open: model weights, card, evaluation results
Class II	Open Tooling	Training code, evaluation code, inference code, and tools
Class I	**Open Science**	Fully open: data, code, parameters, research, documentation

This enables financial institutions to select models based on transparency, auditability, and compliance readiness – not just benchmark performance.

GOVERNANCE

The Financial AI Governance Stack

A layered architecture for safe deployment of AI-powered financial systems.

BUSINESS APPLICATIONS

Financial AI Use Cases

Trading assistants | Risk analysis | Research agents | Advisory systems

AGENT GOVERNANCE — AgentOps

Operational Controls for AI Agents

Decision tracing | Tool monitoring | LLM-as-a-Judge | Human escalation

MODEL GOVERNANCE — FinLLM Benchmarking

Ensuring Models Are Safe to Deploy

Financial benchmarks | Numerical reasoning | Hallucination testing

TRANSPARENCY & COMPLIANCE — Model Openness Framework

Model Provenance and Auditability

Training data transparency | Licensing clarity | Reproducibility

07

What This Means for Banks

Practical governance layers for financial AI deployment

FOR FINANCIAL INSTITUTIONS

Three governance layers for financial AI

Banks deploying GenAI should implement three governance layers.

1

Model Evaluation

Benchmark models against financial datasets. Can the model reason about financial statements? Does it hallucinate in regulatory documents?

2

Agent Monitoring

Track AI agents operating in production. Decision tracing, audit logs, escalation procedures, and runtime evaluation of agent behaviour.

3

Model Transparency

Deploy only models meeting transparency requirements. Training data provenance, licensing clarity, reproducibility, and regulatory auditability.

Financial AI should be treated like regulated financial infrastructure

Benchmarked

Against standardised financial tasks

Monitored

Through operational evaluation in production

Governed

Through compliance and transparency frameworks

Auditable

With traceable model behaviour

This is not consumer AI.

This is financial infrastructure.

Thank You

Questions & Discussion